



Today's agenda

↳ factory design pattern

↳ factory method design pattern

↳ Abstract factory design pattern.



AlgoPrep



SO

<< Database >>



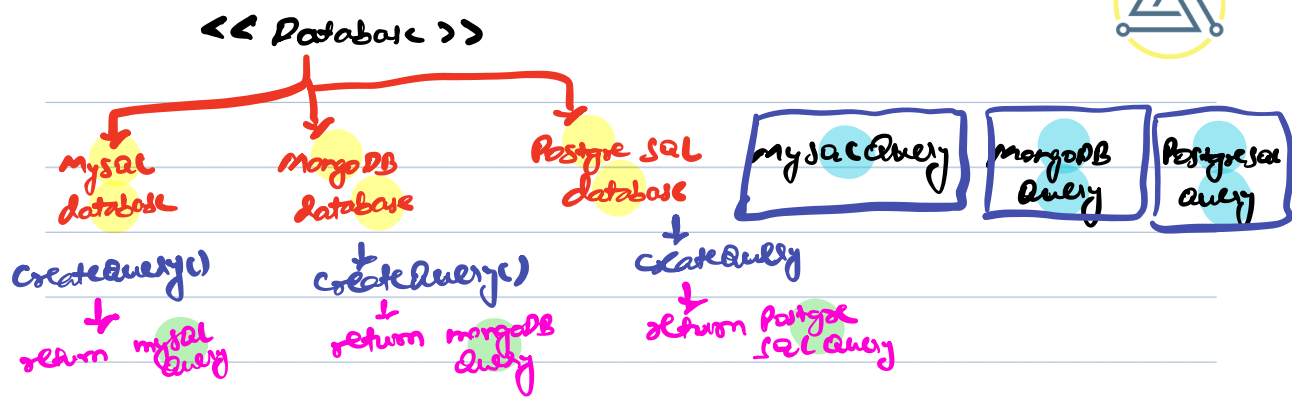
↳ To generate query based on type of database:

Database db = input();

SRP x x
OCP x x

```
if (db instanceof mysql) {  
    createQuery for mysql  
}  
else if (db instanceof mongoDB) {  
    createQuery of mongoDB  
}  
}
```

12



Database db = input();

?? = db.createQuery();

13



Database db = input();

Query q = db.createQuery();

q.execute();



→ Factory method: A method in an interface/abstract class whose purpose is to return the object of a certain class.

* what if database interface has many factory method?

<< Database >>

createQuery();

createConnection();

generateurl();

seturl();

setpassword();

factory method

common method

responsibility of database interface

↳ Has methods needed to perform db operation.

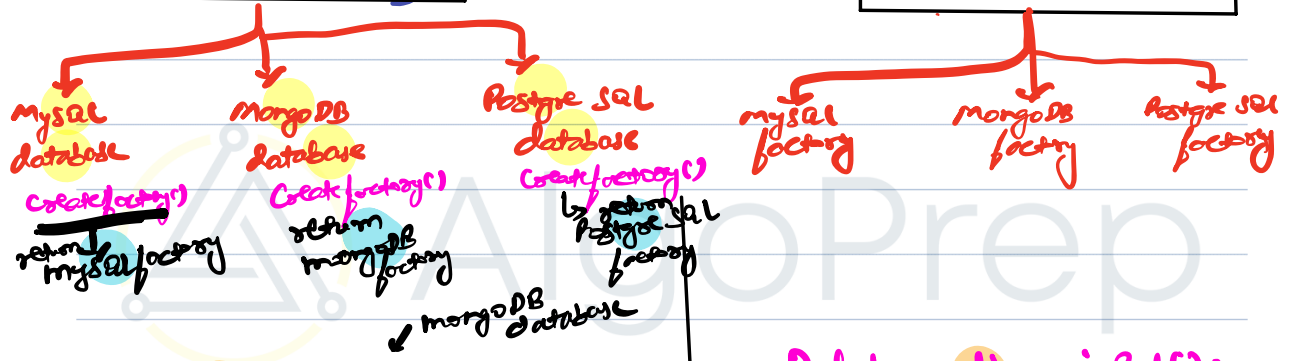
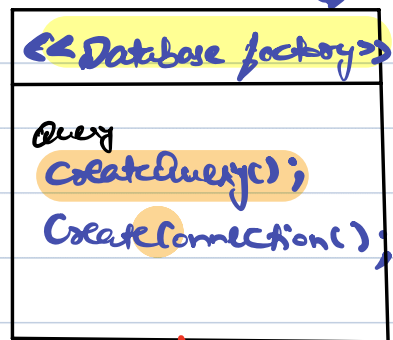
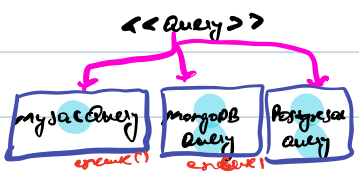
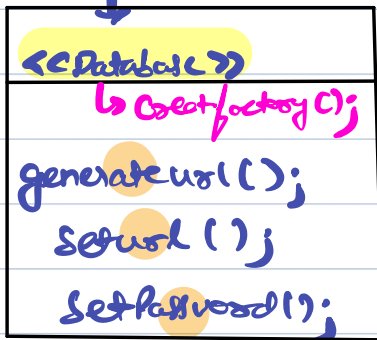
↳ create objects of diff class.

↓
SRP is violated



Segregate the responsibility

abstract factory



Database db = input();

Query q = db.createQuery();

Database db = input();

```
DatabaseFactory dbf;
if (db instanceof MySQL) {
    dbf = new MySQLFactory();
} else if (db instanceof MongoDB) {
    dbf = new MongoDBFactory();
}
...
```

final soln

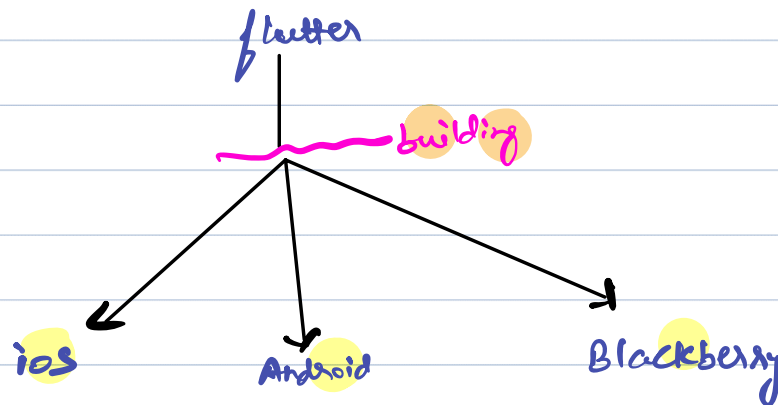
Database db = input();
DatabaseFactory dbf = db.createFactory();

Query q = dbf.createQuery();

Query q = dbf.createQuery();

Create one factory method in database which returns the database factory instance.

Break till 10:20 Pm



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